ASTRA-1

Concept Proposal & Supporting Document
AI-Assisted Adaptive Data Prioritization for Spacecraft

1. Problem Statement

Spacecraft can collect large amounts of camera and sensor data, while communication bandwidth, available energy, and communication opportunities can be limited. Transmitting every piece of data with the same priority may waste valuable communication and power resources. A spacecraft therefore needs a way to decide which information should be transmitted first, especially when an important scientific observation or spacecraft-health event occurs.

2. Proposed Solution

ASTRA-1 proposes an onboard intelligent data-management system that evaluates incoming sensor and camera information and assigns transmission priority. The system will consider factors such as the importance of an observation, spacecraft-health status, communication conditions, and available power or transmission resources. Critical information will receive the highest priority, while routine information can be stored or transmitted later.

3. Core Research Question

Can an adaptive AI-based system dynamically prioritize spacecraft data using scientific value, spacecraft-health information, and communication conditions, while reducing unnecessary data transmission and preserving critical information?

4. Prototype Concept

Subsystem	Purpose
Camera / sensors	Collect visual and environmental/structural information.
Arduino	Interface with sensors and perform low-level measurements.
Raspberry Pi Zero 2 W	Run data processing and the experimental AI/data-prioritization logic.
Communication link	Demonstrate prioritized transmission between the prototype and a ground station.
Structural monitoring	Detect changes in a conductive test structure and generate a health event.
Ground station	Display received data, priority level, alerts, and system status.

5. Experimental Plan

The prototype will create different types of simulated spacecraft data and communication conditions. The system will compare ordinary transmission with adaptive prioritization. Measurements will include the amount of data transmitted, time required to deliver high-priority information, energy/resource use where measurable, and the accuracy of priority classification. The results will be used to determine whether the proposed approach provides a measurable improvement.

6. Expected Application

The concept is intended for spacecraft and other remote autonomous systems where communication bandwidth and energy are constrained. Possible applications include Earth-observation spacecraft, scientific spacecraft, remote sensing platforms, and autonomous robotic systems.

7. Innovation / Research Direction

The project will not claim that onboard data processing or data prioritization is itself a new invention. The research will focus on developing and experimentally evaluating a specific adaptive prioritization method that combines observation importance, spacecraft-health information, and communication/resource conditions. The final novelty claim will be determined only after reviewing existing literature and comparing the experimental method with established approaches.

8. Current Development Status

Idea stage / early concept. The physical prototype, software implementation, experiments, and quantitative results will be developed in subsequent stages.

Note: ASTRA-1 is a ground-based research prototype inspired by spacecraft systems. It is not intended for launch or operation in space as part of this project.